

Variable precision and accuracy

Distinguishes numerical storage precision from actual measurement accuracy. Values below are based on the current repository code and comments; an empirical sensor error distribution is not inferred where the repo does not provide one.

Source: trips.geojson

Scope: Existing classified incidents and stored trip measurements

Variable	Stored / calculated precision	Accuracy / interpretation	Source / caveat
latitude	6 decimal places	~11 cm decimal-place resolution; practical GPS accuracy is much coarser.	Source comment states GPS floor is roughly 1–3 m.
longitude	6 decimal places	~11 cm decimal-place resolution at the equator; practical GPS accuracy is much coarser.	Storage precision is not the same as positional accuracy.
GNSS Speed	Speed capped at 40 km/h and smoothed with a 5-point rolling average; output Speed rounded to 0.1 km/h.	No independent speed error bound is specified in the repo.	Do not interpret 0.1 km/h as sensor accuracy.
Braking intensity	Calculated as (previous speed – current speed) / elapsed time; rounded to 2 decimals; capped at 50 km/h/s.	Resolution of the reported calculation, not an uncertainty estimate.	Threshold for is_braking is 2.0 km/h/s in the current GNSS path.
Peak force / acc_y	acc_y decoded/rounded to 0.001 g in the source processing; crash peak_g exported to 0.01 g.	No accelerometer calibration/error tolerance is specified in the repo.	Reported g values should not be treated as ±0.01 g accurate.
Crash speed at impact	Rounded to 0.1 km/h.	Wheel-rotation estimate; explicitly not GNSS speed. Events whose estimate reaches ≥40 km/h are not trusted by the generator.	Short wheel-rotation lookbacks can be unreliable.
Recovery time	Rounded to 0.01 s.	Represents the gap between wheel-stall readings and the next distinct wheel-rotation reading.	Temporal resolution depends on the underlying data1/sample timing.
Crash onset / suddenness	Rounded to 0.01 s.	Time from onset of the relevant accelerometer excursion to the peak sample.	Raw accelerometer timing uses sample/timestamp mapping in the generator.
timestamp	ISO timestamp on segments; crash point currently exports time_str (HH:MM:SS).	Full crash date/time is reconstructed here only when a same-trip segment timestamp can be matched to the crash clock time.	If no match exists, the PDF retains the original time_str.
Crash coordinates	Point geometry stored as longitude/latitude.	Explicitly approximate.	Current generator writes location_approximate = true.
road_quality	Integer score already calculated in the pipeline.	A categorical/derived road-quality score, not a direct sensor accuracy measurement.	0 means unavailable/unknown in the current frontend.

How to read this table

There are three different concepts here:

1. **storage precision** — how many decimals are written to the file
2. **calculation precision** — how values such as braking_intensity are rounded
3. **measurement accuracy** — how close a sensor-derived value is to the true physical value

The repository provides explicit information for the first two and only limited statements about the third, so this report does not manufacture error margins.

Repository-specific notes

- Coordinate precision is currently COORD_PRECISION = 6; the generator comments explicitly say this is below the GPS accuracy floor.
- GNSS speed uses SPEED_SMOOTH_WIN = 5 before braking calculations.
- The generator caps displayed GNSS speed at MAX_SPEED_KMH = 40 km/h.
- The crash marker location is nearest GNSS position to the crash onset, then carried into the exported Point feature.